AWS CLOUD INFRASTRUCTURE ENGINEERING

Assignment-01-Secure-Compute-Storage-Serverless

SECURE COMPUTE | STORAGE TIERING | SERVERLESS INTEGRATION

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1. Original Assignment Scenario

BUSINESS CONTEXT — A media streaming enterprise is re-architecting its ingestion pipeline. The target architecture requires a secure **Amazon EC2** instance for legacy batch-processing workloads, an **Amazon S3** data lake with lifecycle-based storage tiering, and an **AWS Lambda** function for real-time image thumbnail generation.

TASK 1 — EC2 & SECURITY HARDENING: Design a hardened EC2 environment using custom AMIs, Security Groups that restrict SSH access to specific CIDR blocks, and automatic instance recovery through CloudWatch metric alarms and EC2 Recover.

TASK 2 — S3 STORAGE TIERING STRATEGY: Transition raw video uploads from S3 Standard to Standard-IA after 30 days and S3 Glacier Instant Retrieval after 90 days. Enforce encryption (SSE-S3 or SSE-KMS) and enable versioning.

TASK 3 — SERVERLESS INTEGRATION: Configure an S3 Put event to asynchronously invoke Lambda for thumbnail processing. Address execution timeout, memory allocation, and IAM execution roles.

EXPECTED DELIVERABLE: An architectural specification containing storage lifecycle rules, a Security Group matrix, and a serverless event-flow diagram.

2. Executive Summary

This assignment demonstrates secure cloud infrastructure configuration using **Amazon EC2, Amazon S3, AWS Lambda, IAM** and **Amazon CloudWatch**. The implementation covers hardened compute provisioning, lifecycle-based storage management and event-driven image processing.

A custom Amazon Machine Image (AMI) was created from an updated Amazon Linux 2023 instance and used to launch the final EC2 instance. SSH access was restricted through a Security Group, and a CloudWatch alarm was configured to invoke EC2 Recover for qualifying system-level failures.

The S3 bucket was configured with Block Public Access, versioning, **SSE-S3** encryption, and lifecycle transitions to Standard-IA after 30 days and Glacier Instant Retrieval after 90 days.

A Python 3.12 Lambda function with a Pillow Layer processes JPG objects uploaded under **uploads/** and saves thumbnails under **thumbnails/**. End-to-end JPG upload testing successfully generated a thumbnail.

Implementation boundary: EC2 hardening and S3-to-Lambda thumbnail generation are independent workflows. No legacy batch application or cron job was installed or executed.

3. AWS Services Used

Service	Purpose
Amazon EC2	Virtual compute environment for the legacy batch-processing use case.
AMI	Reusable, preconfigured EC2 image.
Amazon S3	Stores original images and thumbnails; applies lifecycle, encryption and versioning.
AWS Lambda	Processes matching JPG uploads and generates thumbnails.
AWS IAM	Controls Lambda access to S3 objects and CloudWatch logs.
Amazon CloudWatch	Monitors EC2 system status and provides Lambda logs.
AWS CloudShell	Used to package and publish Pillow.
Lambda Layers	Supplies Pillow to the Python 3.12 function.

4. Task 1 — EC2 and Security Hardening

Objective: Prepare a hardened, repeatable EC2 compute environment using a custom AMI, restricted SSH access and automatic instance recovery.

4.1 Custom AMI Creation

A custom AMI named **assignment-01-secure-batch-processing-ami** was created from an updated Amazon Linux 2023 instance. **AMI ID:** ami-07c987294f2bc292b. The AMI was Available at verification. The temporary AMI-builder instance was terminated after image creation. An AMI is a reusable image template, not a running instance.

4.2 EC2 Provisioning

The final instance **assignment-01-secure-batch-processor** was launched from the custom AMI. **Instance type:** t3.micro. It passed status checks during verification. The legacy batch-processing application, script and cron schedule were not installed or implemented.

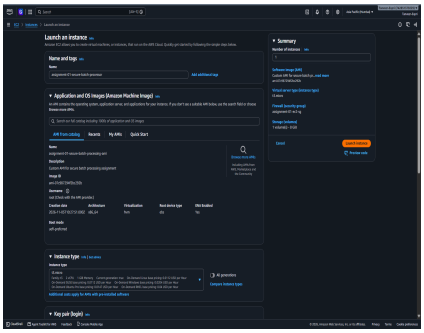
4.3 Security Group Configuration

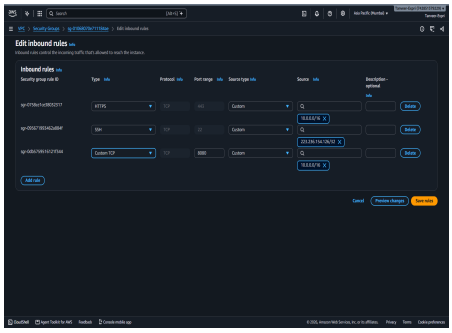
Type	Protocol	Port	Source
SSH	TCP	22	223.236.154.126/32
HTTPS	TCP	443	10.0.0.0/16
Custom TCP	TCP	8080	10.0.0.0/16

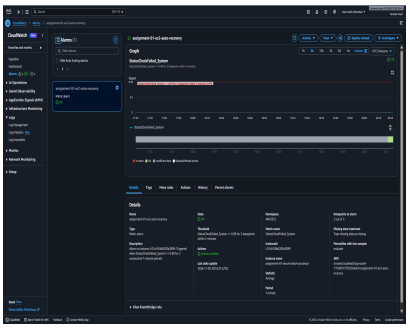
SSH is restricted to the specified public IP. HTTPS and TCP 8080 allow traffic from the VPC CIDR. **Confirm that 10.0.0.0/16 matches the actual VPC CIDR** before applying these rules.

4.4 CloudWatch Automatic Recovery

Alarm: assignment-01-ec2-auto-recovery. **Metric:** StatusCheckFailed_System. **Evaluation:** two one-minute periods. **Action:** EC2 Recover. This supports recovery for qualifying underlying host failures; it does not restart an instance intentionally stopped or terminated by a user.







5. Task 2 — Amazon S3 Storage Tiering

5.1 Bucket Configuration

Bucket: tanveer-assignment-01-media-datalake. **Region:** ap-south-1 (Mumbai). Configured features: Block Public Access, versioning, SSE-S3 encryption and lifecycle management.

5.2 Lifecycle Policy

Lifecycle rule: assignment-01-video-lifecycle. Configured transitions:

Storage class	Transition time
S3 Standard	Initial storage class
S3 Standard-Infrequent Access (Standard-IA)	After 30 days
S3 Glacier Instant Retrieval	After 90 days

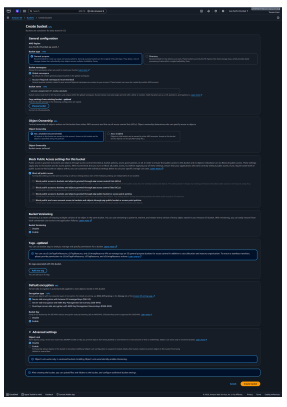
The rule applies to the entire bucket. A production design may use a **videos/** prefix if images, thumbnails and videos need different retention policies. Minimum storage duration, retrieval charges and object-size considerations can affect savings.

5.3 SSE-S3 Encryption

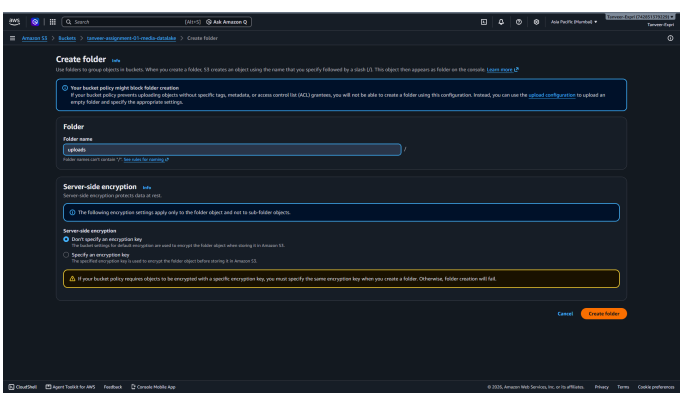
Server-side encryption is enabled using **SSE-S3**, which uses S3-managed keys. SSE-KMS is an alternative with additional key-management controls and audit capabilities, but may introduce KMS request charges. **Implemented:** SSE-S3. **Not implemented:** SSE-KMS.

5.4 S3 Versioning

S3 Versioning was enabled. It retains multiple versions of an object and can help recover from accidental overwrites or deletions. Retained versions consume storage and can increase costs.







6. Task 3 — Serverless Thumbnail Generation

6.1 Lambda Function

Function: assignment-01-thumbnail-generator. **Runtime:** Python 3.12. **Architecture:** x86_64. **Memory:** 512 MB. **Timeout:** 30 seconds. **Ephemeral storage:** 512 MB.

Memory provides working memory and affects available CPU proportionally. Timeout sets the maximum execution duration. Ephemeral storage provides temporary disk space during invocation.

6.2 IAM Least-Privilege Access

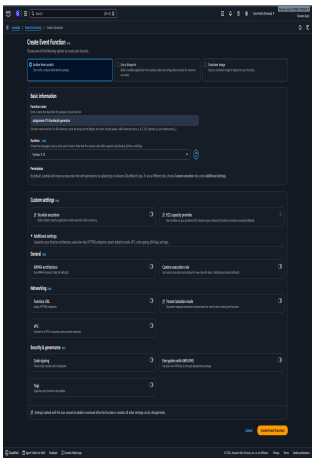
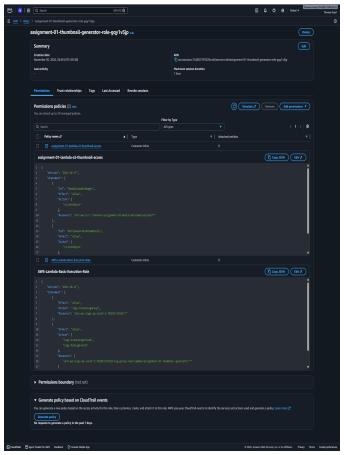
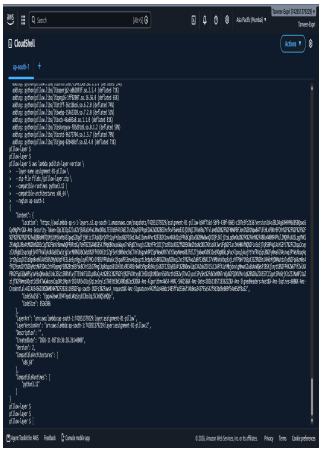
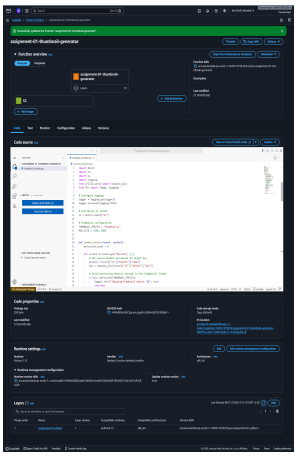
Inline policy: assignment-01-lambda-s3-thumbnail-access. It grants **s3:GetObject** on uploads/* and **s3:PutObject** on thumbnails/* in the assignment bucket, plus basic CloudWatch logging permissions.

Permission	Resource
s3:GetObject	arn:aws:s3:::tanveer-assignment-01-media-datalake/uploads/*
s3:PutObject	arn:aws:s3:::tanveer-assignment-01-media-datalake/thumbnails/*

6.3 Pillow Layer Creation

Pillow was packaged using AWS CloudShell and published as a Lambda Layer. CloudShell used Python 3.13 while Lambda used Python 3.12, so a Python 3.12-compatible Linux x86_64 package was downloaded using pip platform and ABI options.

Layer: assignment-01-pillow **Version:** 1
ARN: arn:aws:lambda:ap-south-1:742851379229:layer:assignment-01-pillow:1

7. Deployment, Testing and Architecture Overview

7.1 S3 Event Notification

The S3 trigger listens for ObjectCreated events on the assignment bucket, filtered by prefix **uploads/** and suffix **.jpg**. Matching uploads asynchronously invoke Lambda. The output prefix **thumbnails/** differs from the input prefix, avoiding recursive invocation.

7.2 Lambda Processing Logic

The function decodes the S3 object key, downloads the image, corrects EXIF orientation, resizes it to fit within **200 × 200 pixels**, converts it to RGB when necessary, encodes it as JPEG, and writes the thumbnail to **thumbnails/**.

Input: uploads/Butterfly.jpg **Output:** thumbnails/Butterfly.jpg

7.3 Deployment and End-to-End Testing

The Python code was deployed and the Pillow Layer attached successfully. A JPG uploaded under **uploads/** triggered Lambda, which generated and saved the thumbnail under **thumbnails/**. The test succeeded.

7.4 CloudWatch Logs

CloudWatch Logs provide visibility into Lambda execution and image-processing activity.

Architecture Summary

Task 1 — Secure Compute: Custom AMI → EC2; Security Group → restricted SSH and VPC-only ingress; CloudWatch alarm → EC2 Recover.

Task 2 — Storage Tiering: S3 with Block Public Access, versioning and SSE-S3; Standard → Standard-IA after 30 days → Glacier Instant Retrieval after 90 days.

Task 3 — Serverless: User upload → S3 uploads/ → ObjectCreated event → Lambda + Pillow → S3 thumbnails/.

Important: EC2 is not part of thumbnail generation. Stopping EC2 does not interrupt S3-to-Lambda processing while the bucket, trigger, Lambda and permissions remain configured.

8. Conclusion

The assignment demonstrates secure EC2 provisioning, restricted network access, CloudWatch recovery configuration, S3 lifecycle transitions, SSE-S3 encryption, versioning, IAM least-privilege access and successful event-driven thumbnail generation.

Implementation limitations: The legacy batch application was not deployed. SQS and SSE-KMS were not implemented. The lifecycle rule applies to the entire bucket rather than a video-only prefix.

